Stationary theory versus Langevin-annealed training ($T_0 = 10^{-5}$, momentum 0, 50k steps, seed 0)

f_C: readout-weighted compositional share; R: accessibility (cyan $R = 1/\sqrt{8}$); φ_{out}: block-2 2+2→4 share of the readout-weighted $d4$ amplitude; dashed: theory H_C region

lower objective: per cell, the run with the lower final objective J (W = parent-warm, J = joint)

Theory: phase

Theory: f_C

Theory: R

Theory: φ_{out}

Parent-warm + anneal: phase

Parent-warm + anneal: f_C

Parent-warm + anneal: R

Parent-warm + anneal: φ_{out}

Joint + anneal: phase

Joint + anneal: f_C

Joint + anneal: R

Joint + anneal: φ_{out}

Lower objective: phase

Lower objective: f_C

Lower objective: R

Lower objective: φ_{out}

block-1 decay ratio λ_1/λ_2

common decay scale λ_0

Stationary theory versus Langevin-annealed training ($T_0 = 10^{-7}$, momentum 0, 150k steps, seed 0)

f_C: readout-weighted compositional share; R: accessibility (cyan $R = 1/\sqrt{8}$); φ_{out}: block-2 2+2→4 share of the readout-weighted $d4$ amplitude; dashed: theory H_C region

lower objective: per cell, the run with the lower final objective J (W = parent-warm, J = joint)

Theory: phase

Theory: f_C

Theory: R

Theory: φ_{out}

Parent-warm + anneal: phase

Parent-warm + anneal: f_C

Parent-warm + anneal: R

Parent-warm + anneal: φ_{out}

Joint + anneal: phase

Joint + anneal: f_C

Joint + anneal: R

Joint + anneal: φ_{out}

Lower objective: phase

Lower objective: f_C

Lower objective: R

Lower objective: φ_{out}

block-1 decay ratio λ_1/λ_2

common decay scale λ_0

Stationary theory versus deterministic training (momentum 0.9, 150k steps, seed 0)

f_C: readout-weighted compositional share; R: accessibility (cyan $R = 1/\sqrt{8}$); φ_{out}: block-2 $2{+}2 \to 4$ share of the readout-weighted $d4$ amplitude; dashed: theory H_C region

lower objective: per cell, the run with the lower final objective J (W = parent-warm, J = joint)

Theory: phase

Theory: f_C

Theory: R

Theory: φ_{out}

Parent-warm: phase

Parent-warm: f_C

Parent-warm: R

Parent-warm: φ_{out}

Joint: phase

Joint: f_C

Joint: R

Joint: φ_{out}

Lower objective: phase

Lower objective: f_C

Lower objective: R

Lower objective: φ_{out}

block-1 decay ratio λ_1/λ_2

common decay scale λ_0

Stationary theory versus deterministic training (momentum 0.9, 150k steps, hidden lr 0.001, readout init var 1/N, readout lr 0.001, seed 0)

f_C: readout-weighted compositional share; R: accessibility (cyan $R = 1/\sqrt{8}$); φ_{out}: block-2 $2{+}2 \to 4$ share of the readout-weighted $d4$ amplitude; dashed: theory H_C region

lower objective: per cell, the run with the lower final objective J (W = parent-warm, J = joint)

Theory: phase

Theory: f_C

Theory: R

Theory: φ_{out}

Parent-warm: phase

Parent-warm: f_C

Parent-warm: R

Parent-warm: φ_{out}

Joint: phase

Joint: f_C

Joint: R

Joint: φ_{out}

Lower objective: phase

Lower objective: f_C

Lower objective: R

Lower objective: φ_{out}

block-1 decay ratio λ_1/λ_2

common decay scale λ_0